PIX2PIX FOR ARCHITECTURAL FACADE IMAGE TRANSLATION: IMPLEMENTATION AND ABLATION STUDY

Student Project Report

Abstract—This project presents a reduced-scale PyTorch implementation of Pix2Pix, a conditional generative adversarial network for translating architectural label maps into facade photographs. A U-Net generator is trained with a conditional PatchGAN discriminator using an adversarial objective and an L1 reconstruction term. The Facades dataset split used here contains 400 training, 100 validation, and 106 test examples. The full model and an ablation without the L1 term were each trained for 50 epochs. On the test set, the full model obtains MAE 0.1847 and PSNR 12.57 dB, while the adversarial-only variant obtains MAE 0.2446 and PSNR 10.29 dB. The results indicate that, under this setup, the L1 term improves pixel-level similarity. Visual inspection shows that both models retain substantial blurring and artifacts; this is a small-scale reproduction rather than a state-of-the-art result.

Keywords—Pix2Pix, conditional GAN, image-to-image translation, U-Net, PatchGAN, Facades, ablation study

I. INTRODUCTION

Image-to-image translation learns a mapping from an input image representation to an output image representation. In architectural facade translation, a semantic label map specifies the spatial arrangement of facade elements, while the desired output is a corresponding photograph. This task provides a useful setting for conditional image synthesis because the label map describes layout but not all appearance details.

Pix2Pix formulates paired image translation as a conditional generative adversarial network (cGAN). A generator produces an output conditioned on the input, and a discriminator evaluates whether an input-output pair resembles a real paired example. The original method combines adversarial learning with an L1 reconstruction term [1]. This project implements the core idea in PyTorch, evaluates the output on a Facades test split, and studies the effect of removing the L1 term.

This is a reduced-scale educational reproduction, not an exact replication of every setting in the original paper. The implementation uses a U-Net with 32 base channels and trains for 50 epochs.

II. METHOD AND ARCHITECTURE

A. Paired Translation. Let x denote a facade label map and y its paired real photograph. The generator G predicts G(x). The conditional discriminator D receives the concatenation of x with either y or G(x). The goal is to generate a photograph that is conditioned on the label map and visually plausible.

B. U-Net Generator. The generator uses an encoder-decoder U-Net design. The encoder progressively downsamples the 256 × 256 input using stride-2 convolutions and LeakyReLU activations. The decoder upsamples using transposed convolutions, batch normalization, and ReLU activations. Skip connections concatenate encoder and decoder features at matching resolutions to preserve spatial layout. The final Tanh activation matches images normalized to [−1, 1]. The generator has three input channels, three output channels, 32 base channels, and approximately 13.61 million parameters.

C. Conditional PatchGAN. The discriminator receives six channels: the three-channel label map concatenated with the three-channel target or generated photograph. Its convolutional network produces a 30 × 30 grid of logits for a 256 × 256 pair, judging local patches rather than assigning a single score to the entire image. It has approximately 0.695 million parameters. Real pairs are assigned target 1 and generated pairs target 0.

III. OBJECTIVE AND TRAINING

The full generator objective combines adversarial and reconstruction losses: $L_G = L_{GAN}(G, D) + \lambda L1(G)$, with $\lambda = 100$. The discriminator objective is $L_D = 0.5[BCE(D(x,y),1) + BCE(D(x,G(x)),0)]$. Binary cross-entropy with logits is used for the adversarial terms and mean absolute error for the L1 term. The reconstruction term penalizes pixel-wise differences to the paired target; the adversarial term encourages outputs that the discriminator considers realistic.

Training settings were: 256 × 256 resolution, batch size 1, 50 epochs, Adam optimizer, learning rate 0.0002, and betas (0.5, 0.999). The data loader normalized images to [−1, 1].

Ablation. A separate generator and discriminator were initialized and trained for 50 epochs with the adversarial term only, $L_G = L_{GAN}(G,D)$. The original checkpoints were kept separate. The comparison investigates whether removing explicit pixel reconstruction supervision changes test performance.

IV. DATASET AND EVALUATION

The experiment uses the paired Facades dataset. Each paired image is split into a label map input and its corresponding real photograph target. The recorded split contains 400 training images, 100 validation images, and 106 test images. Quantitative evaluation is performed on the 106 test examples.

MAE is calculated after converting generated and target tensors from [−1, 1] to [0, 1]. Lower MAE indicates smaller average absolute pixel error. PSNR is computed from mean squared error with peak intensity 1.0; higher PSNR indicates lower pixel reconstruction error. These metrics quantify pixel similarity but do not fully measure perceptual realism or semantic correctness.

V. RESULTS

Table I summarizes test-set performance. The original Pix2Pix model achieves MAE 0.184699 and PSNR 12.5658 dB. The adversarial-only ablation achieves MAE 0.244560 and PSNR 10.2875 dB. Relative to the ablation, the original model reduces MAE by approximately 24.49%. Its PSNR is approximately 2.28 dB higher.

These values support the conclusion that the L1 term improved pixel-level reconstruction under this implementation and training setup. They do not establish that L1 always improves every translation task or that the implementation matches the original paper's reported performance.

VI. QUALITATIVE ANALYSIS

Visual comparison of three selected test examples shows that both models recover broad facade structure and repeated window arrangements, but the outputs remain blurry and contain texture artifacts. In the displayed examples, the original model appears closer to the targets overall, while the adversarial-only outputs show more pronounced distortion and darker or less consistent regions. This observation is consistent with the quantitative metrics, but three examples cannot establish visual superiority across every test image.

The outputs should not be described as high-fidelity synthesis. Their quality suggests room for improvement in training stability, training duration, model configuration, and evaluation.

VII. LIMITATIONS AND FUTURE WORK

Only one training run per configuration is reported, so variability across random seeds is unknown. The study uses a reduced-scale implementation and a single dataset split. MAE and PSNR do not fully capture perceptual quality, and both models show visible artifacts. Although a validation split was prepared, no validation-based checkpoint selection is claimed here.

Future work should repeat experiments with multiple random seeds, report variation, investigate longer training and learning-rate scheduling, use validation-driven checkpoint selection, and include perceptual metrics or structured human evaluation. The code, checkpoints, histories, figure, and CSV results should be packaged with reproducible instructions.

VIII. CONCLUSION

This project implemented a paired facade translation pipeline using a U-Net generator and conditional PatchGAN discriminator. After 50 epochs, the full adversarial-plus-L1 model achieved MAE 0.1847 and PSNR 12.57 dB on 106 test images. A separately trained adversarial-only ablation achieved MAE 0.2446 and PSNR 10.29 dB. The comparison suggests that L1 reconstruction loss improved pixel-level similarity in this experiment. However, visible blurring and artifacts remain, and further controlled experiments are needed.

REFERENCES

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TABLE I. TEST-SET COMPARISON

Model	MAE ↓	PSNR (dB) ↑	L1 used
Pix2Pix (GAN + L1)	0.184699	12.5658	Yes
Ablation (GAN only)	0.244560	10.2875	No